

horizontal velocity is constant and hence, z remains constant.

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B

$$V_{\text{rms}} = \frac{150}{\sqrt{2}}$$

$$P = \frac{V_{\text{rms}}^2}{R}$$

$$13 = \frac{150^2}{2R}$$

$$R = 865 \, \Omega, \text{ hence } R_{\text{total}} = 2 \times 865$$

$$P = \frac{240^2}{2 \times 865} = 33 \, \text{W}$$

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C